

# Appendix A1 - South-west Corner Marine Park (Western arm) Natural Values Report

South-west Marine Parks Network

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## Natural Values

Understanding of natural values

## Sampling summary

Table A1.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

| Date         | Campaign name                          | Method | Areas                         | Samples | Habitat | Fish  | Length |
|--------------|----------------------------------------|--------|-------------------------------|---------|---------|-------|--------|
|              |                                        |        |                               |         | Count   | Count |        |
| Jun 2020     | 2020-06 south-west stereo-BRUVs        | BRUV   | NPZ                           | 42      | 40      | 39    | 33     |
| Oct-Nov 2020 | 2020-10 south-west stereo-BRUVs        | BRUV   | Coastal waters, NPZ, SPZ      | 275     | 273     | 272   | 244    |
| Oct-Nov 2020 | 2020-10 south-west BOSS                | BOSS   | Coastal waters, NPZ, SPZ      | 262     | 262     | -     | -      |
| Mar 2021     | 2021-03 West-Coast BOSS                | BOSS   | Coastal waters, NPZ           | 154     | 154     | -     | -      |
| Mar 2023     | 2023-03 Augusta stereo-BOSS            | BOSS   | Coastal waters, NPZ, SPZ      | 73      | 70      | -     | -      |
| Mar 2023     | 2023-03 SwC BOSS                       | BOSS   | NPZ                           | 88      | 88      | -     | -      |
| Mar 2023     | 2023-03 SwC stereo-BRUVs               | BRUV   | NPZ                           | 32      | 32      | 32    | 32     |
| Oct 2024     | 2024-10 SWC stereo-BRUVs               | BRUV   | Coastal waters, NPZ, SPZ      | 190     | 190     | 188   | 178    |
| Nov 2024     | 2024-11 SwC stereo-BOSS                | BOSS   | Coastal waters, MUZ, NPZ, SPZ | 137     | 117     | -     | -      |
| Mar-Apr 2025 | 2025-04 South-west Corner BOSS         | BOSS   | NPZ, SPZ                      | 210     | 210     | -     | -      |
| Apr 2025     | 2025-04 South-west-corner stereo-BRUVs | BRUV   | NPZ, SPZ                      | 150     | 148     | 143   | 137    |

## Benthic ecosystem extent

### Observations

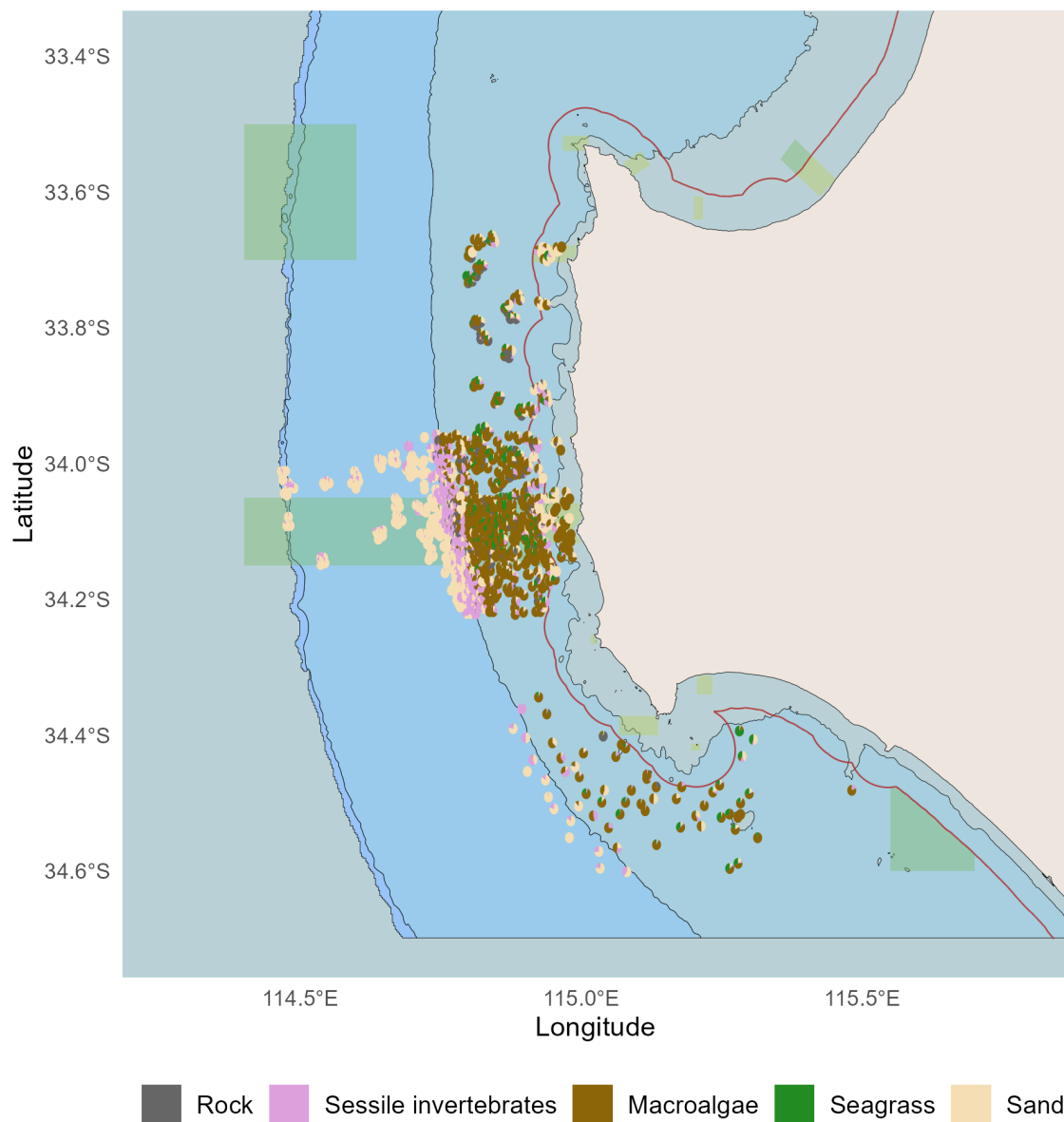


Figure A1.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Dominant habitat types

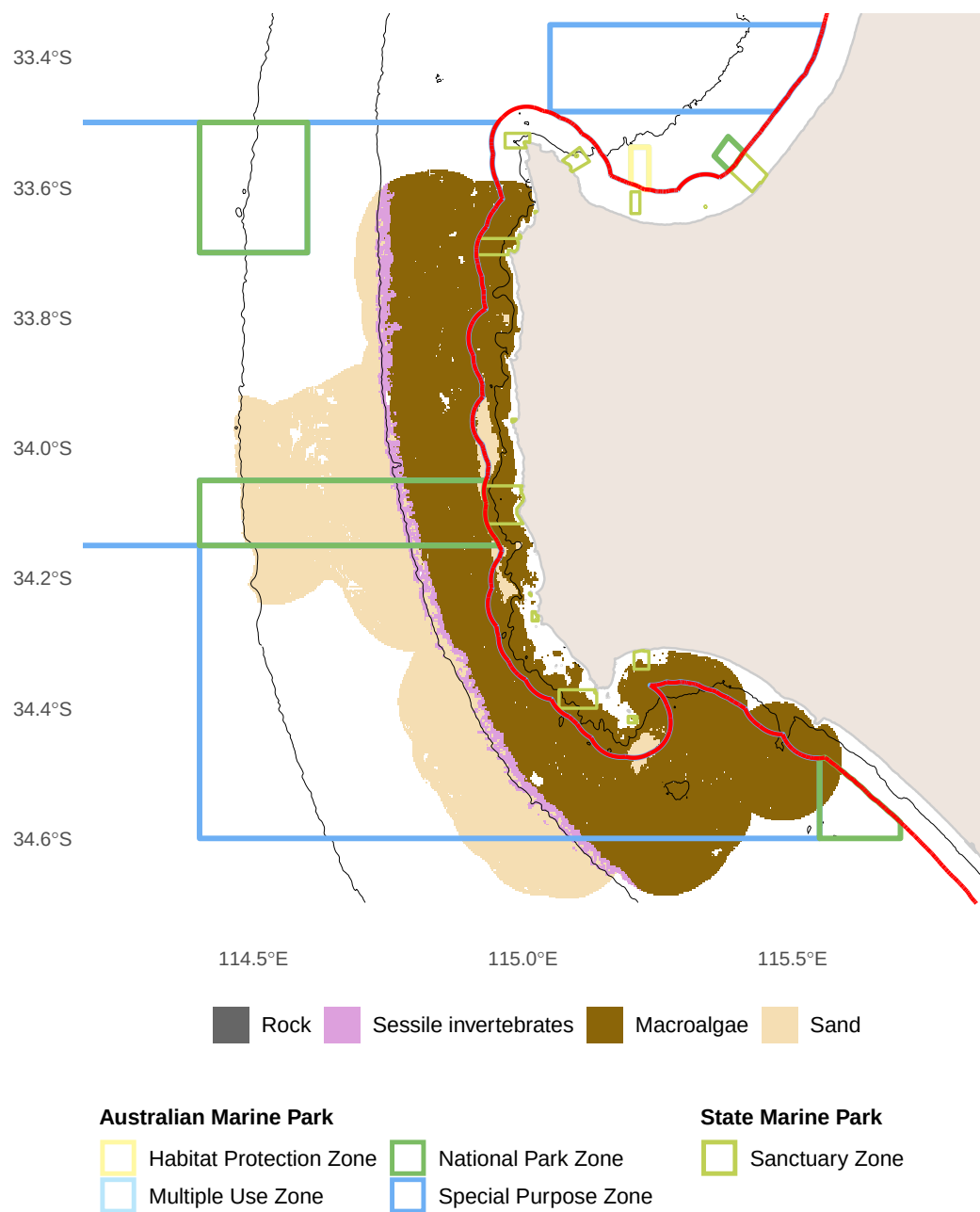


Figure A1.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.



## Habitat distribution and probability

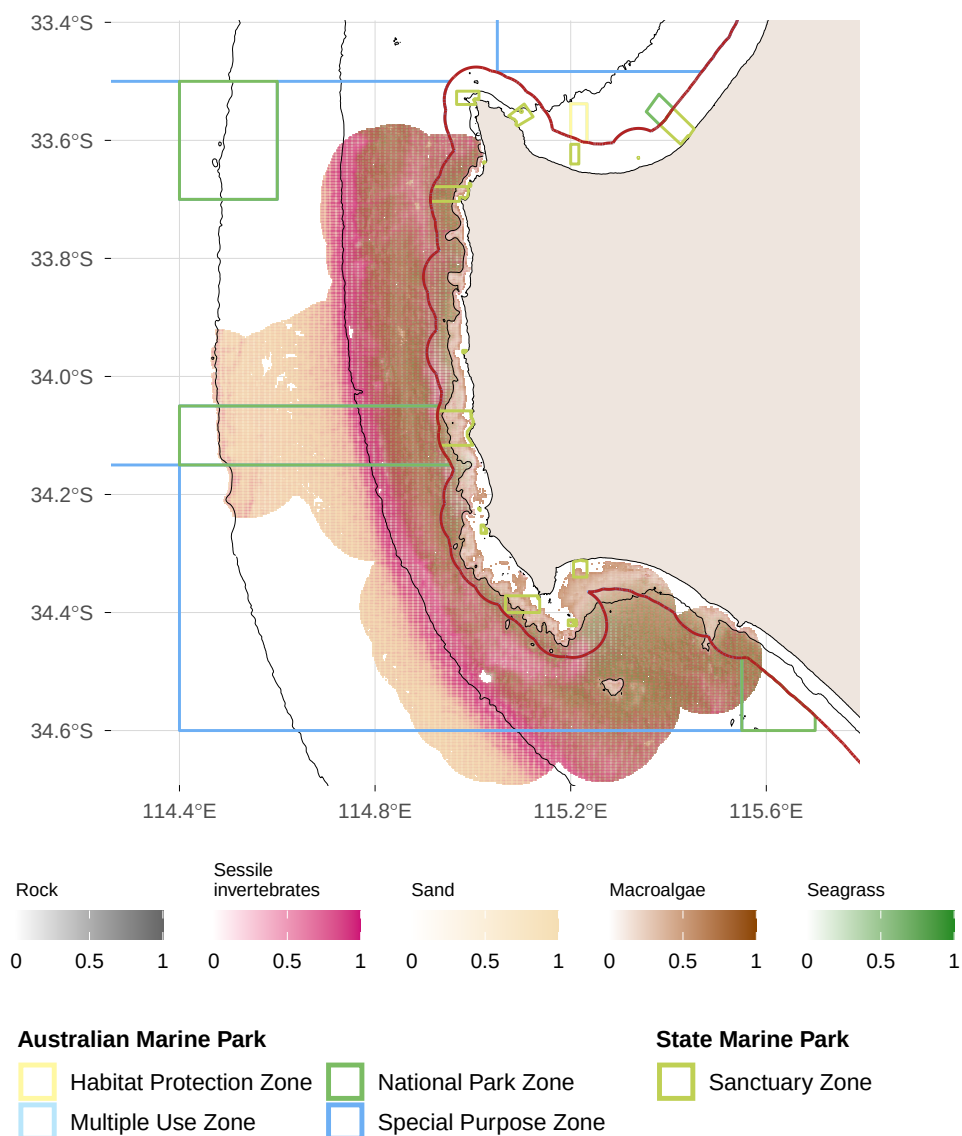


Figure A1.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Functional reef distribution and probability

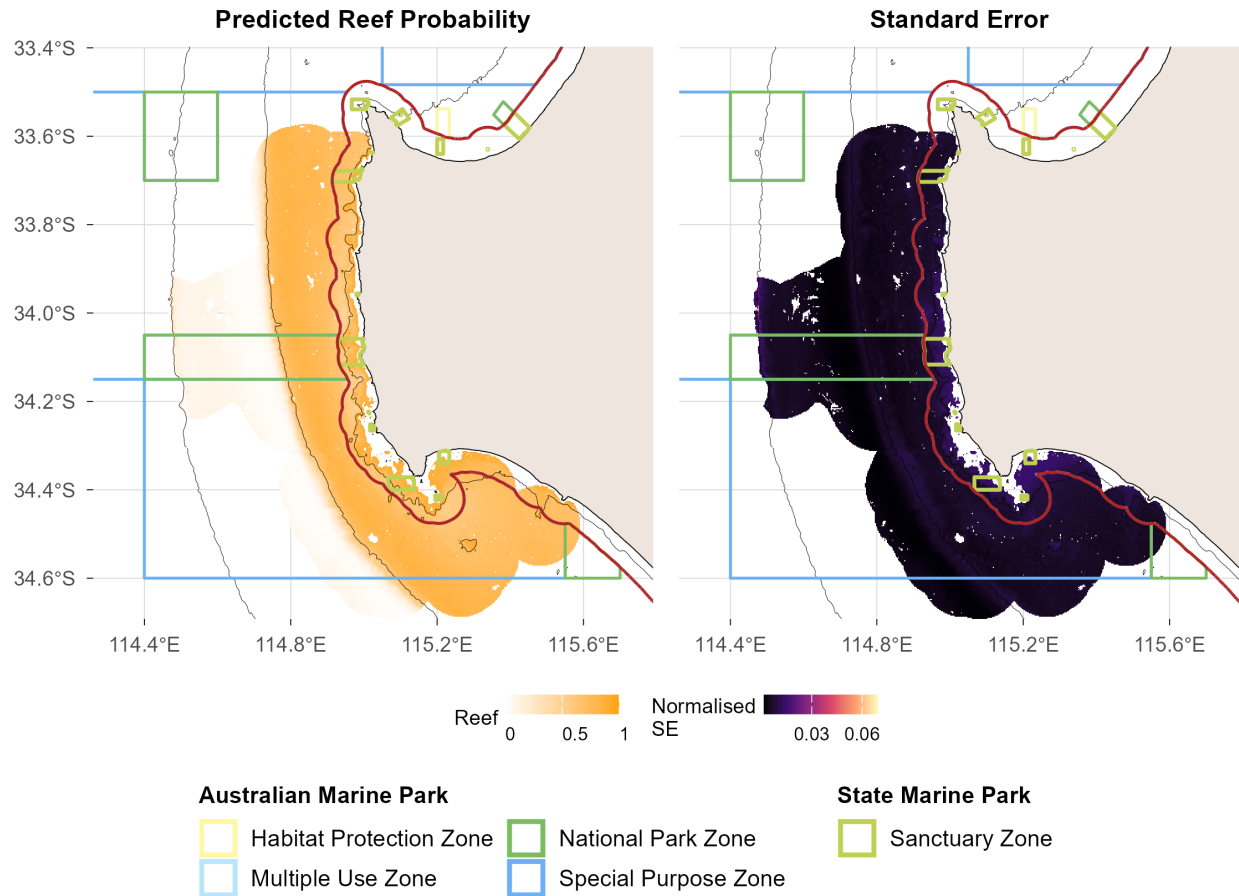


Figure A1.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Benchmarks of benthic natural values

### Sand

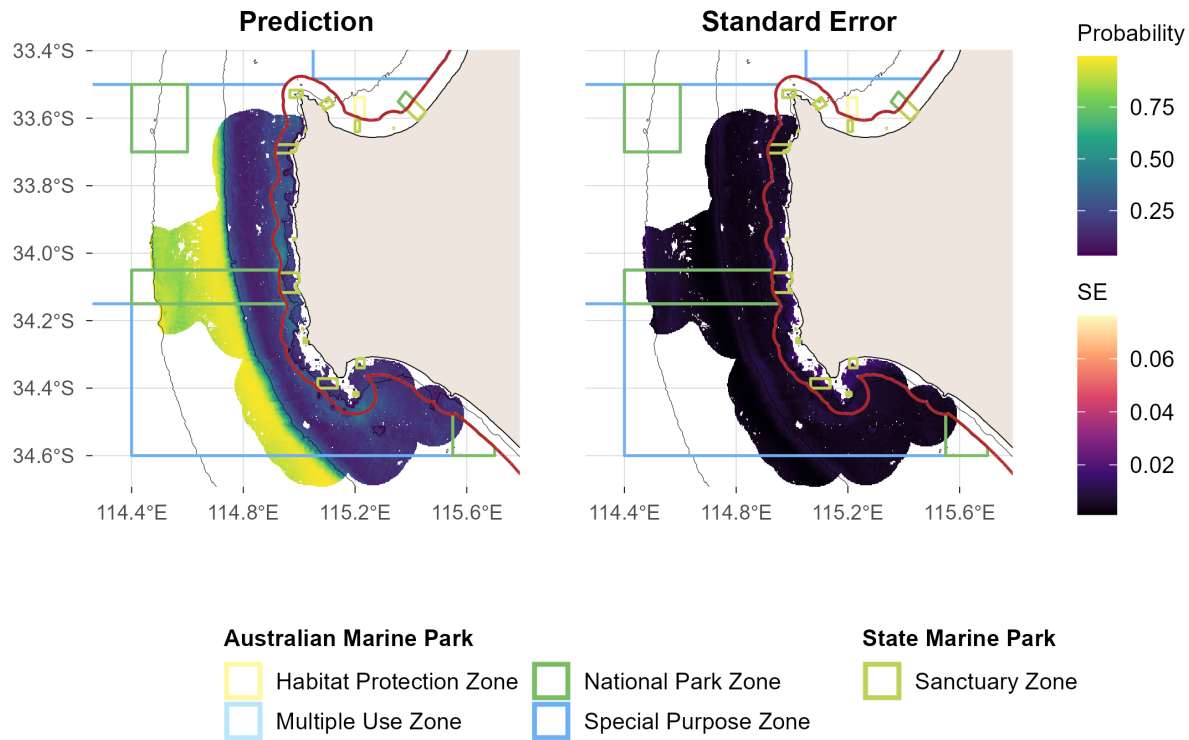


Figure A1.5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Macroalgae

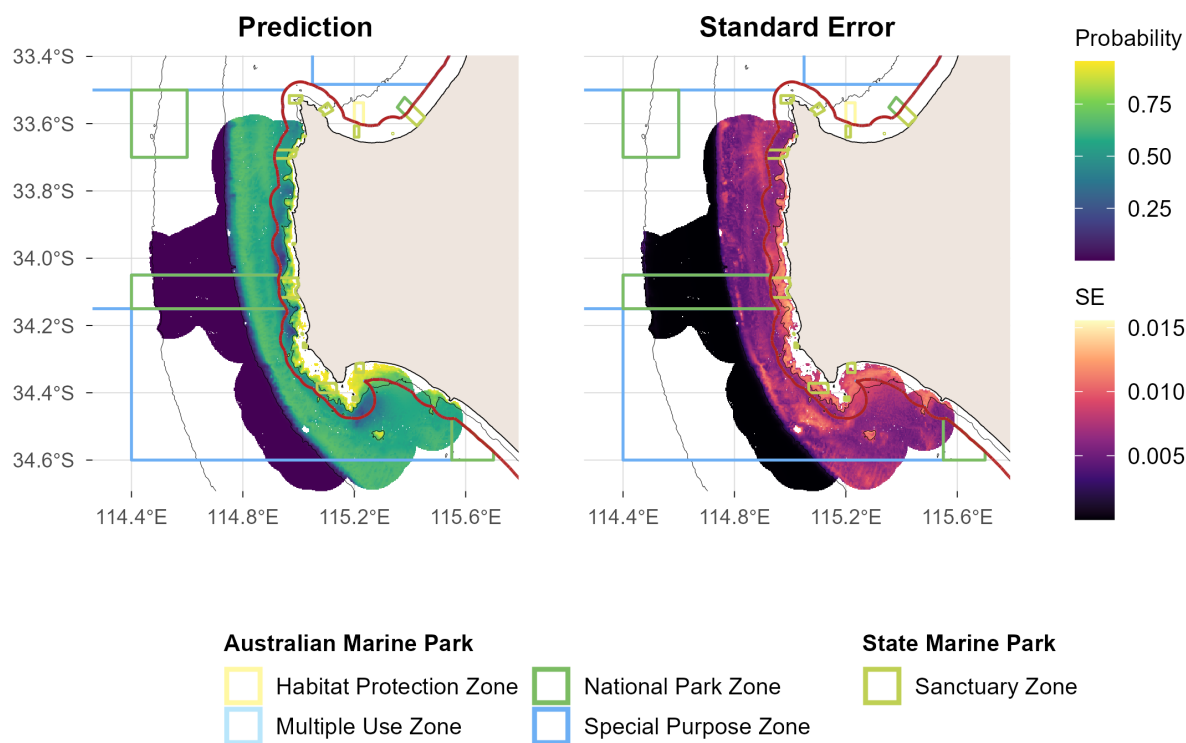


Figure A1.6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Seagrass

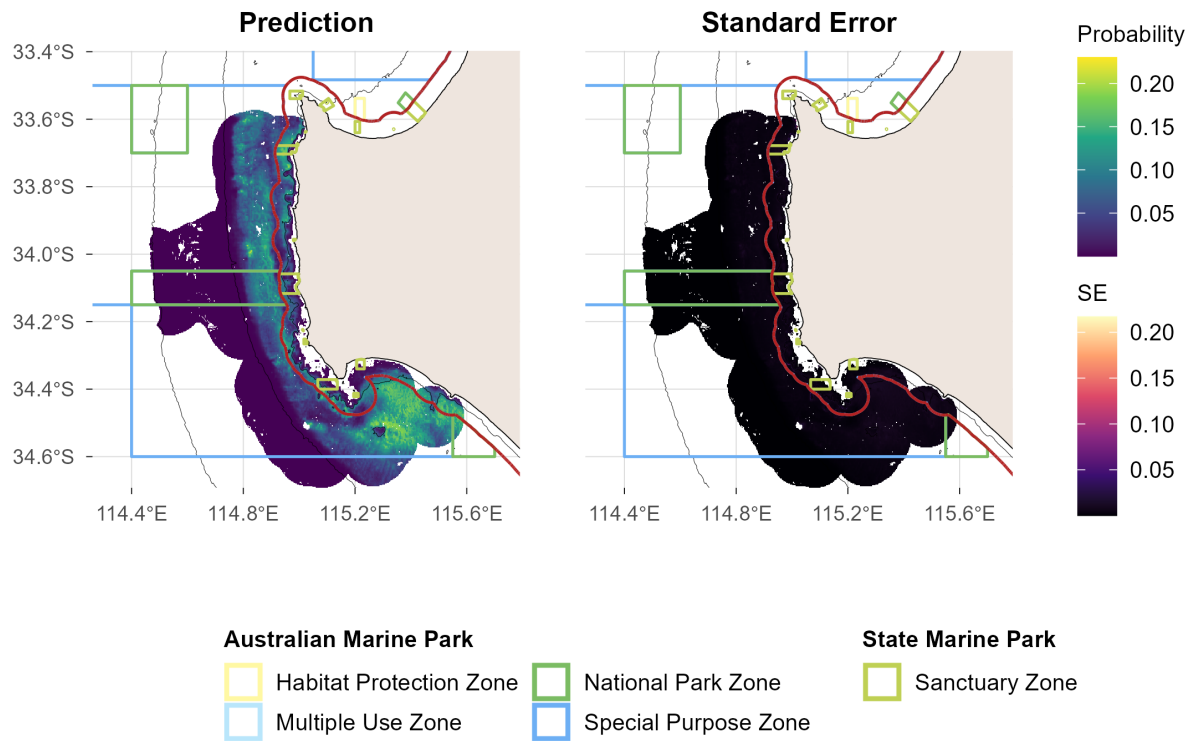


Figure A1.7.1: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Sessile invertebrates

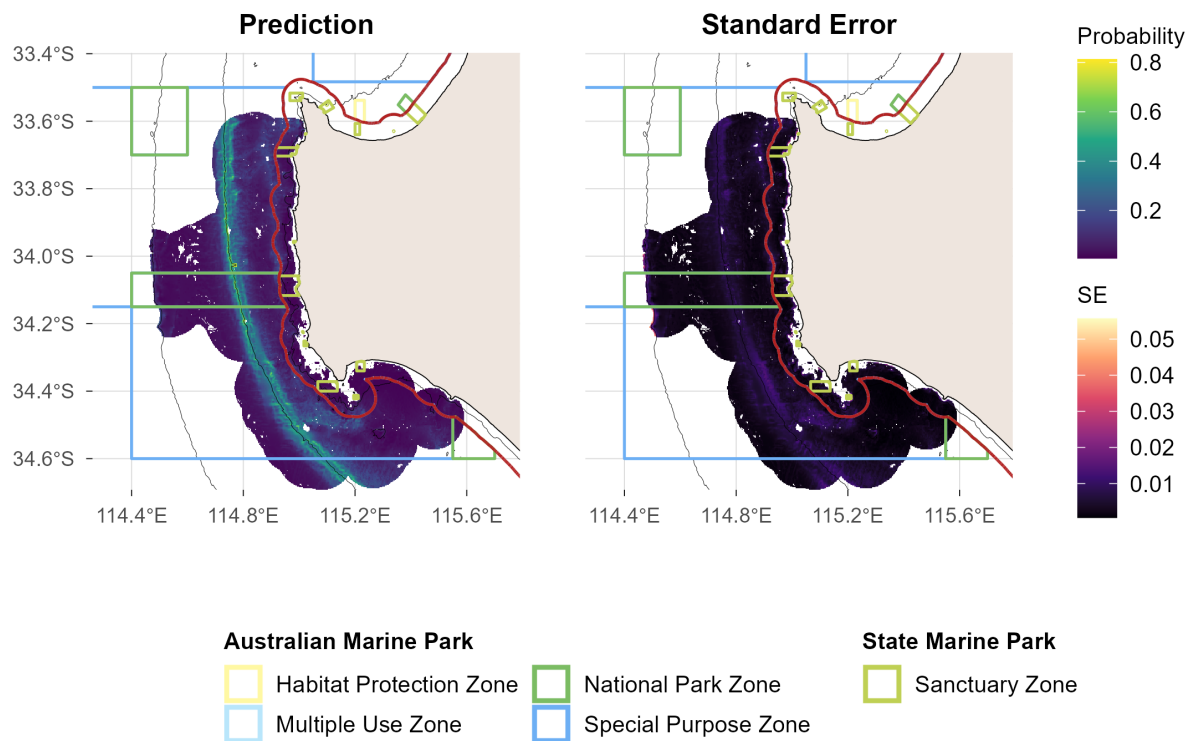


Figure A1.8.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Rock

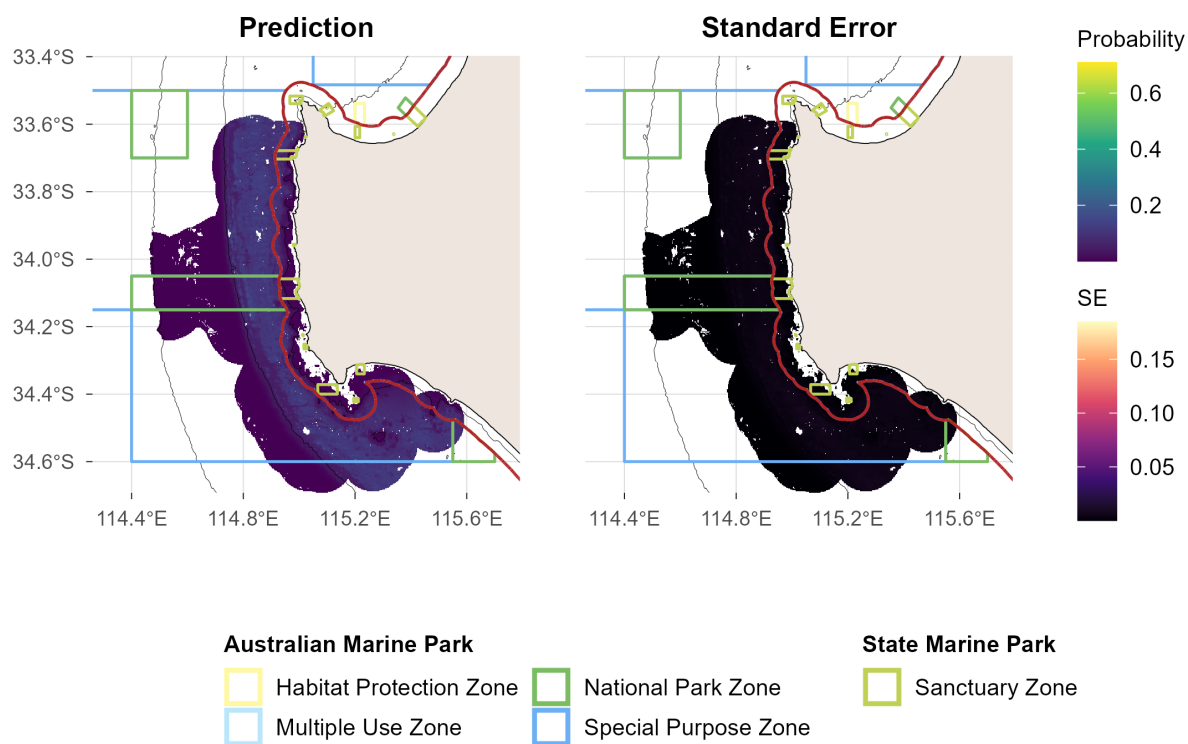
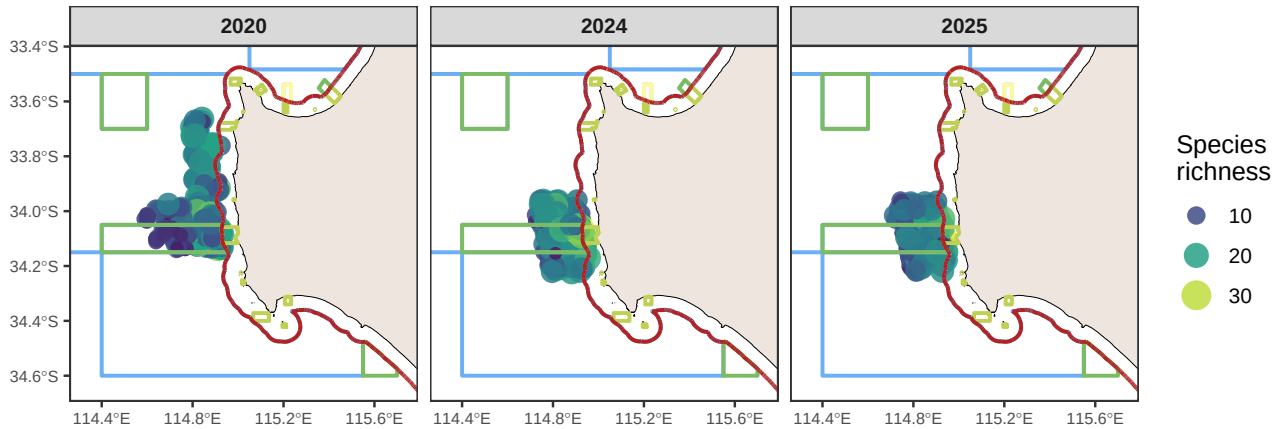


Figure A1.9.1: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

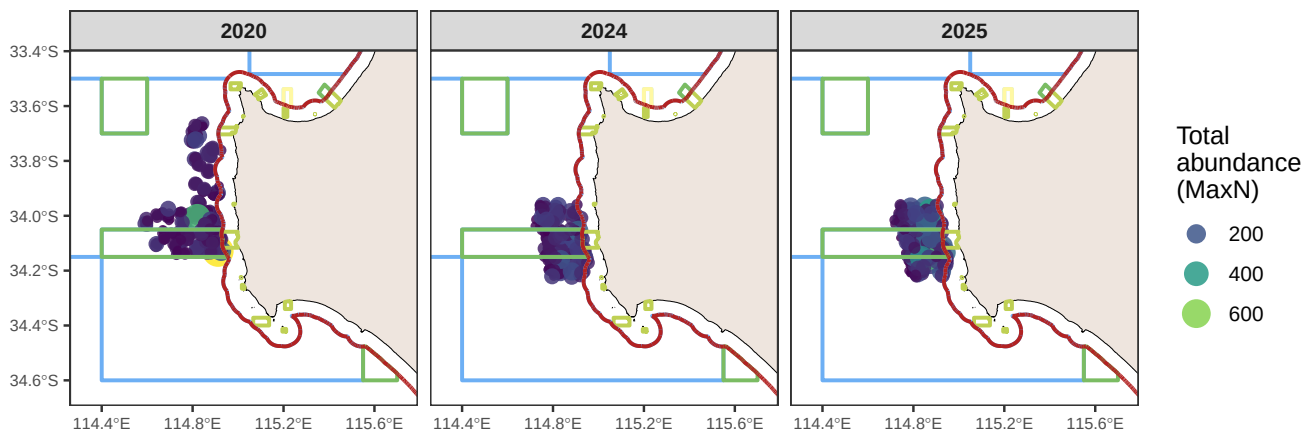
## Benchmarks of demersal fish natural values

### Survey effort

a)



b)



#### Australian Marine Park

Habitat Protection Zone  
Multiple Use Zone

National Park Zone  
Special Purpose Zone

#### State Marine Park

Sanctuary Zone

Figure A1.10: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.



## Species richness

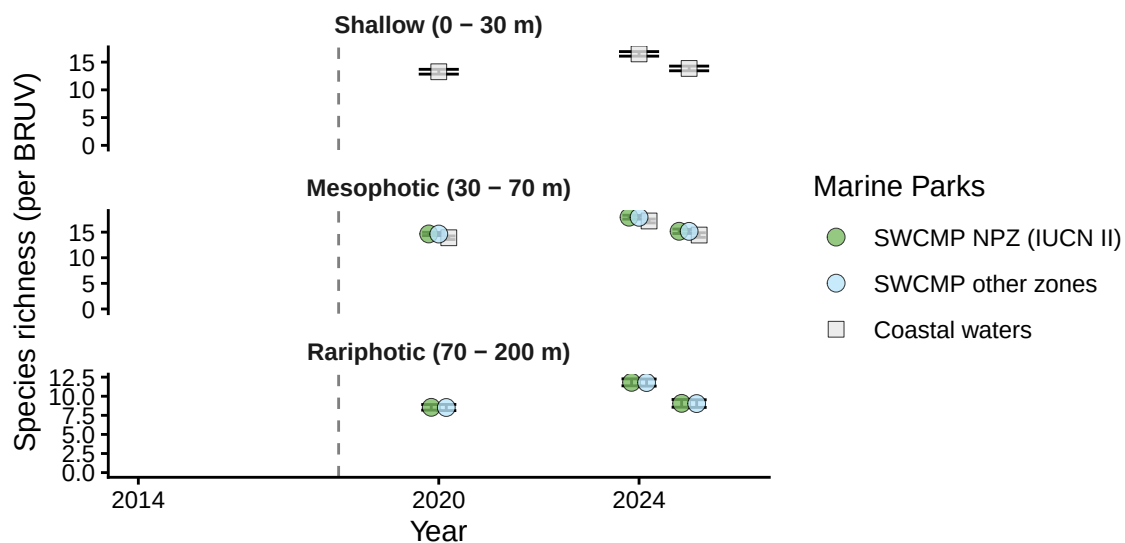


Figure A1.11.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

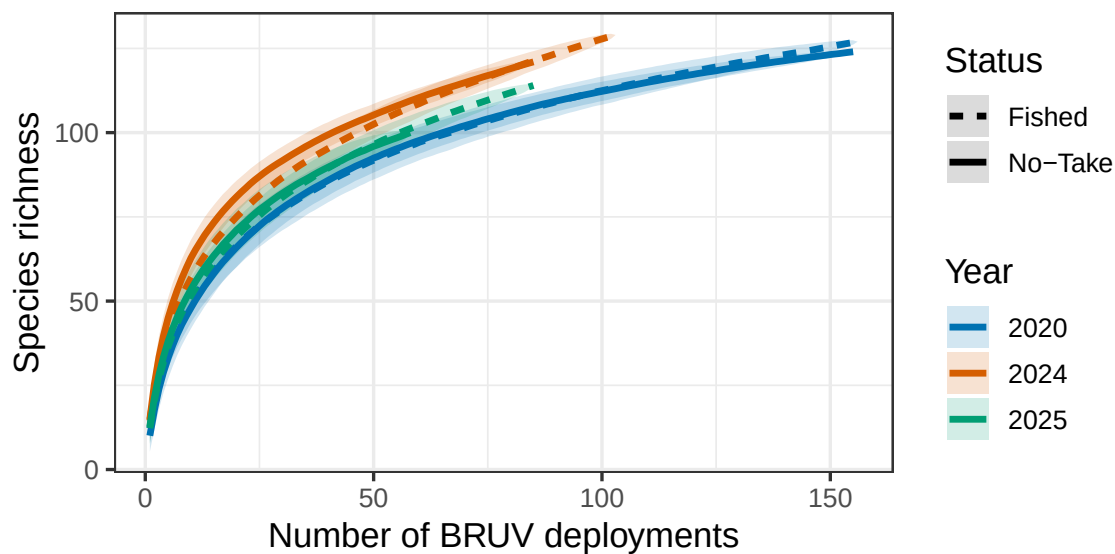


Figure A1.11.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Species richness

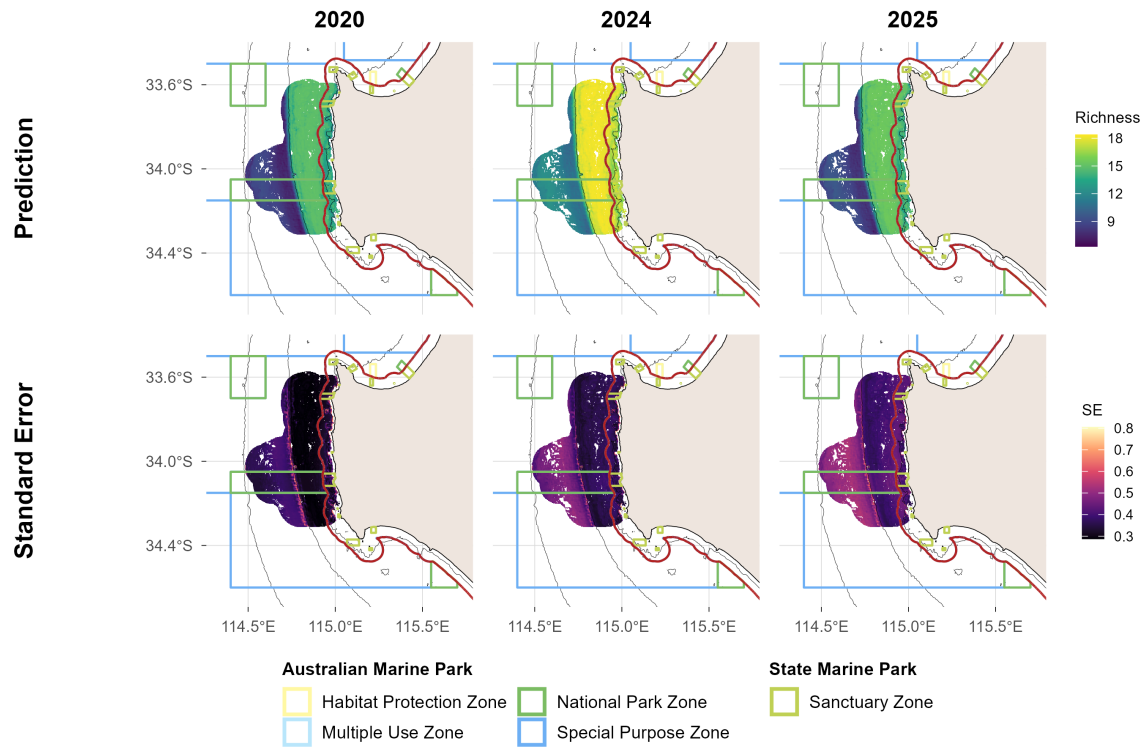


Figure A1.11.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

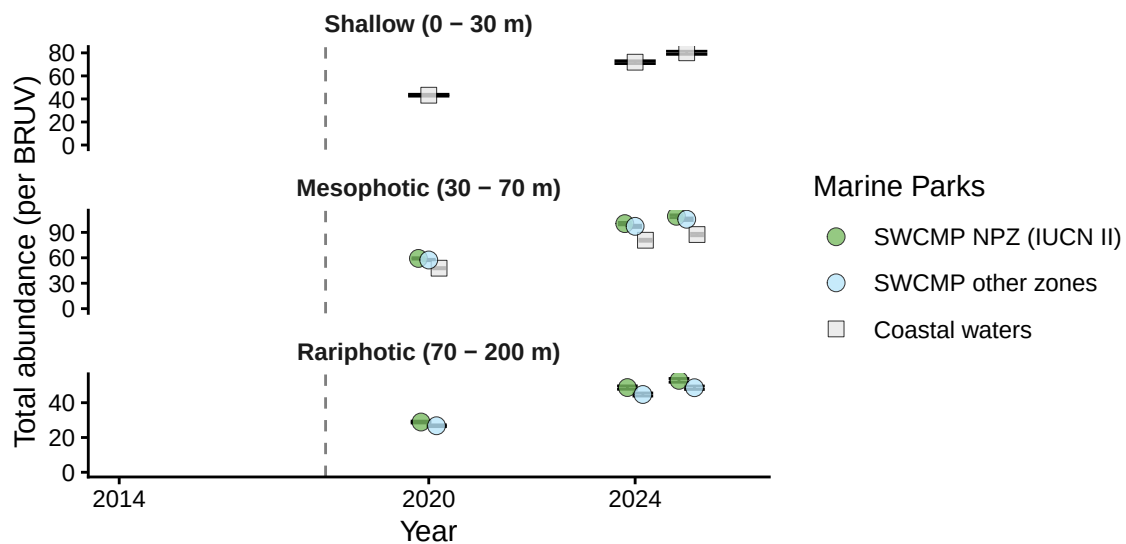


Figure A1.12.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

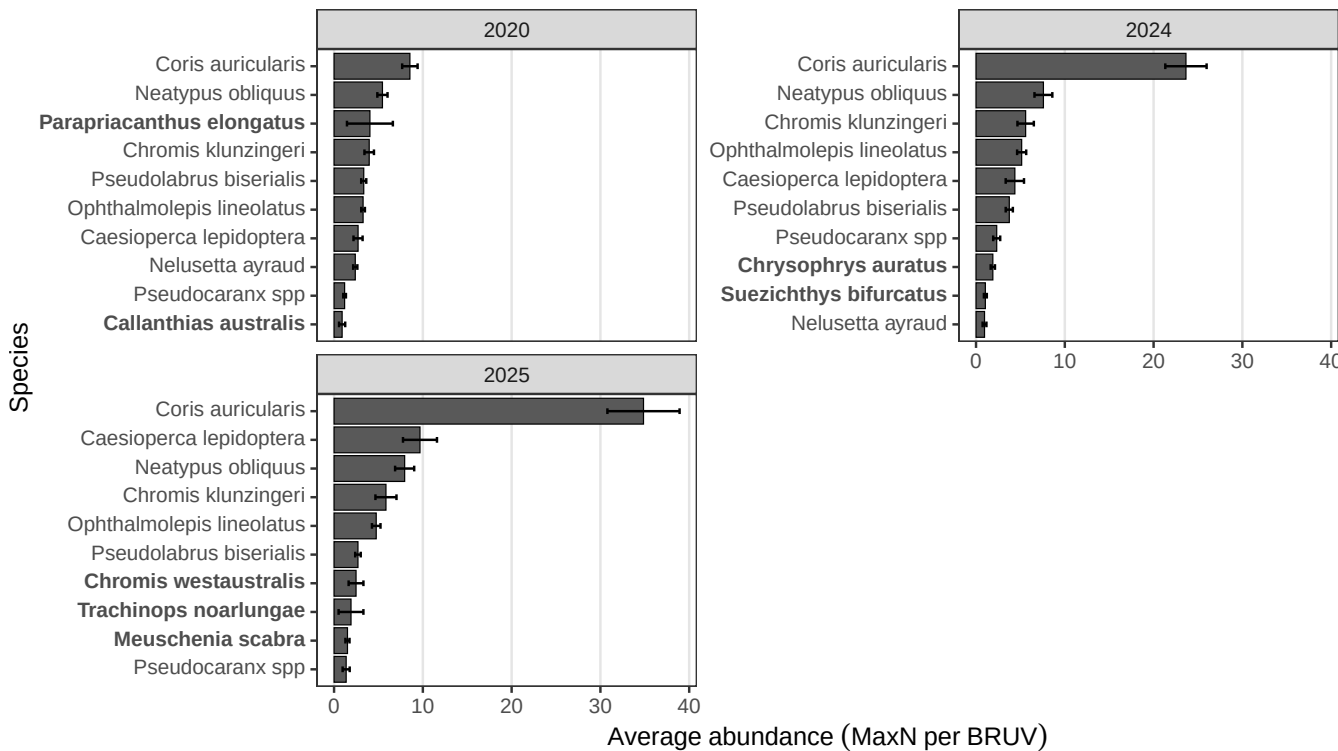


Figure A1.12.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Total abundance

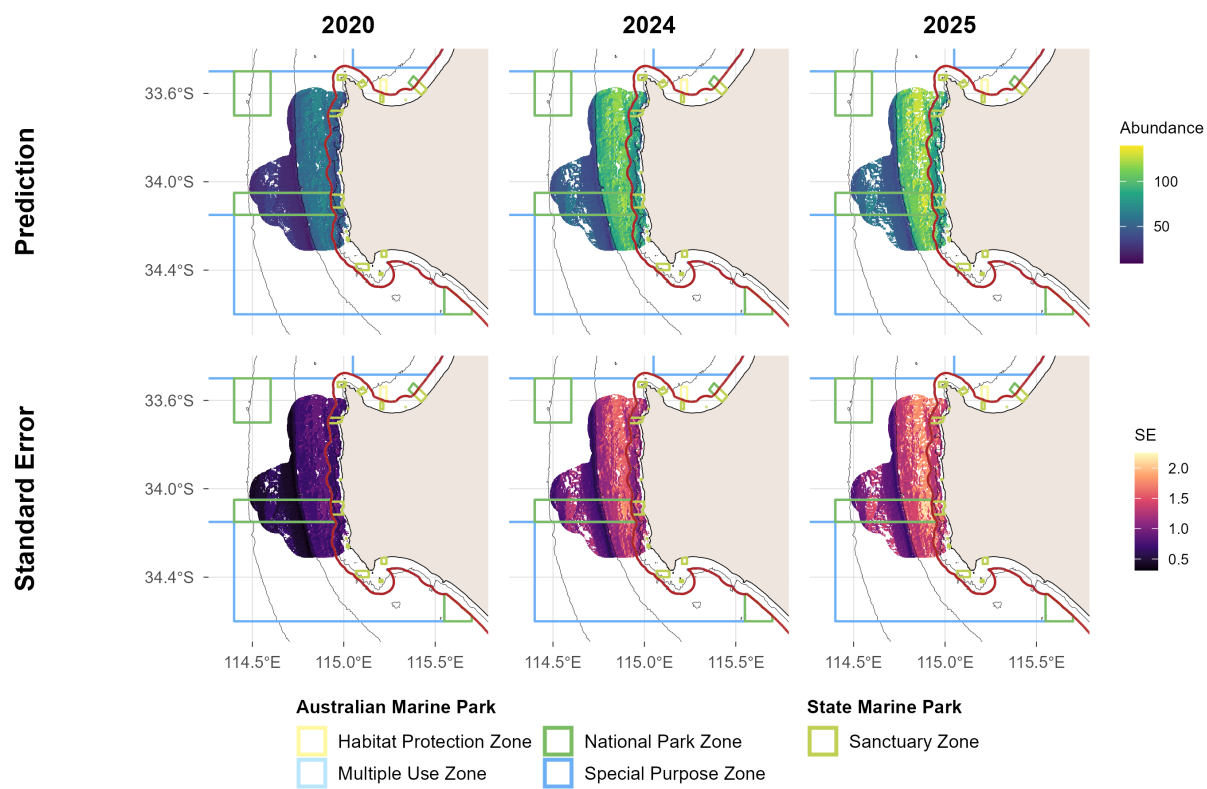


Figure A1.12.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index\*

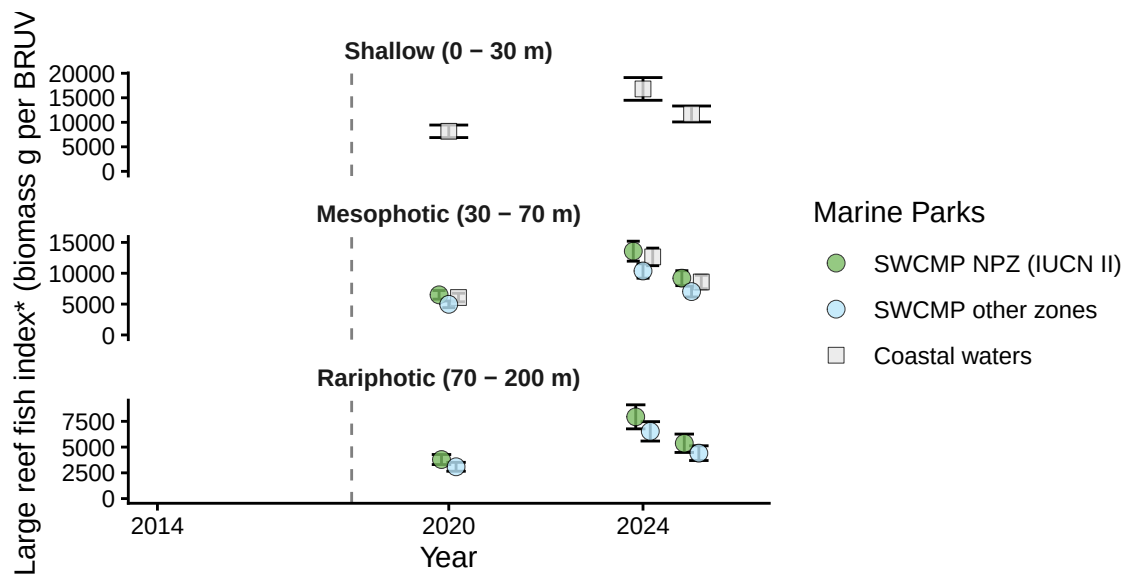


Figure A1.13.1: Demersal fish - Large Reef Fish Index\*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). \*Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

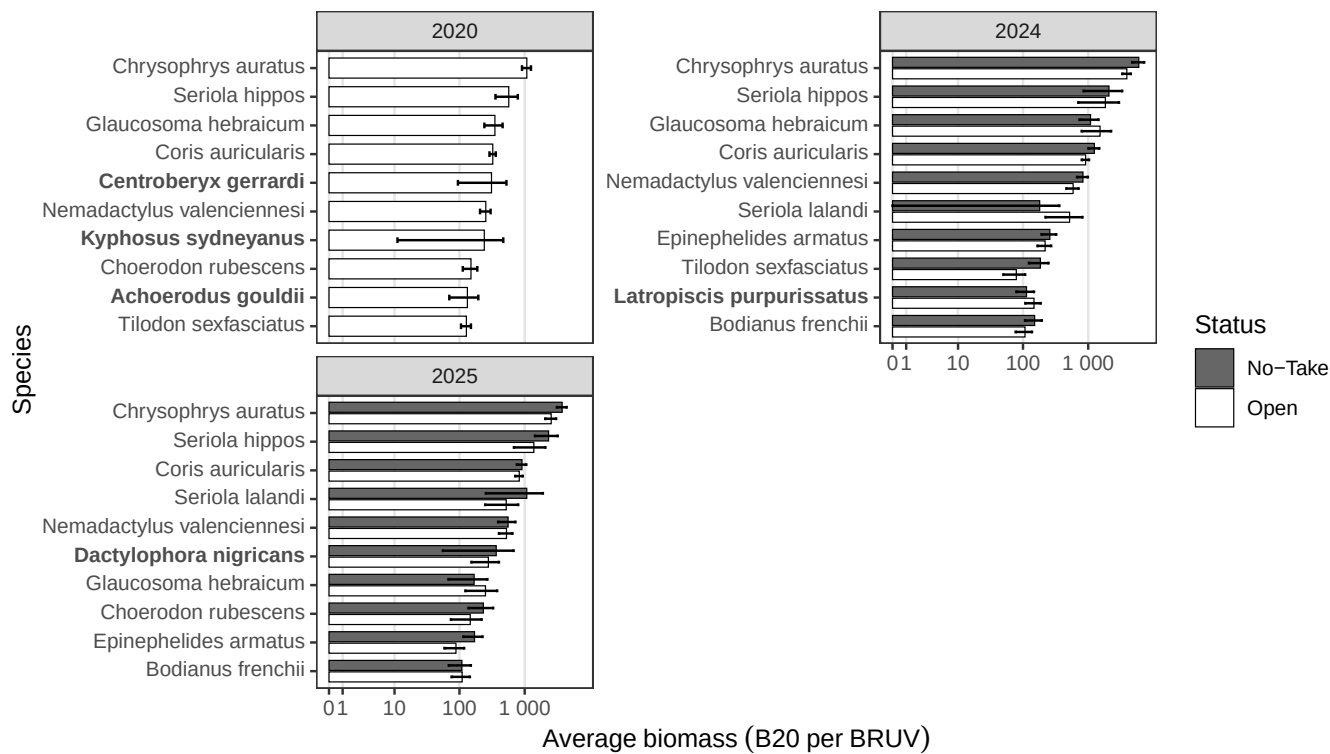


Figure A1.13.2: Demersal fish - Large Reef Fish Index\*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Large Reef Fish Index\*

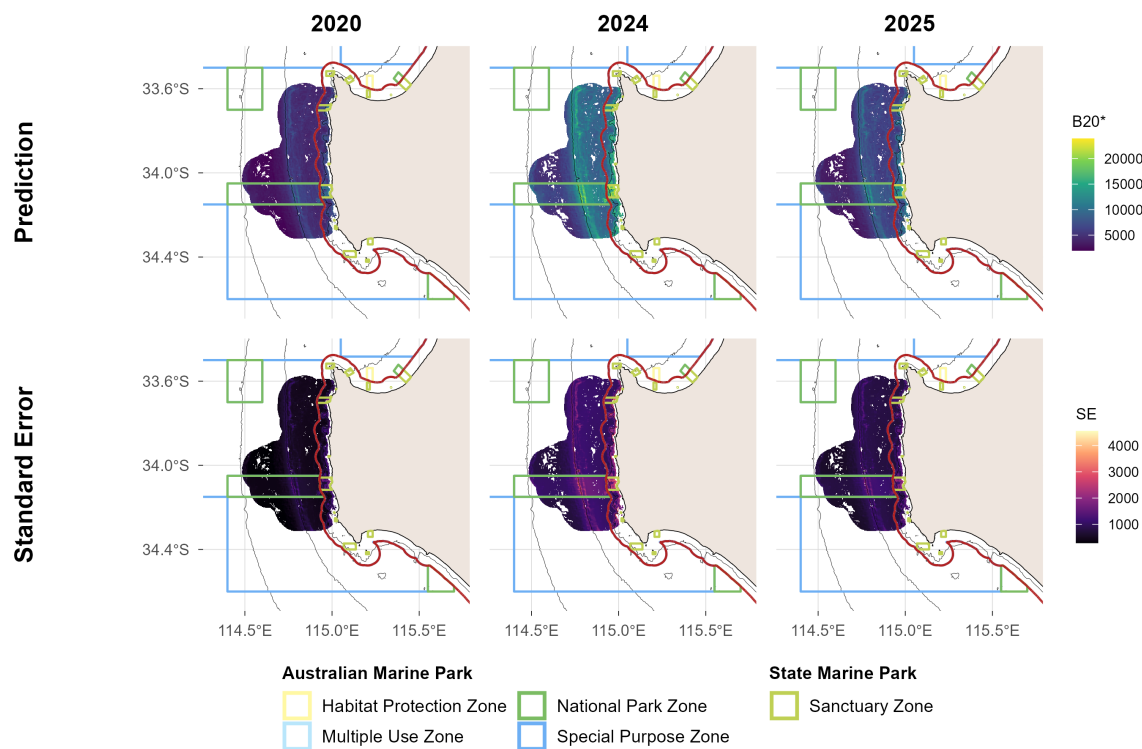


Figure A1.13.3: Demersal fish - Large Reef Fish Index\*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

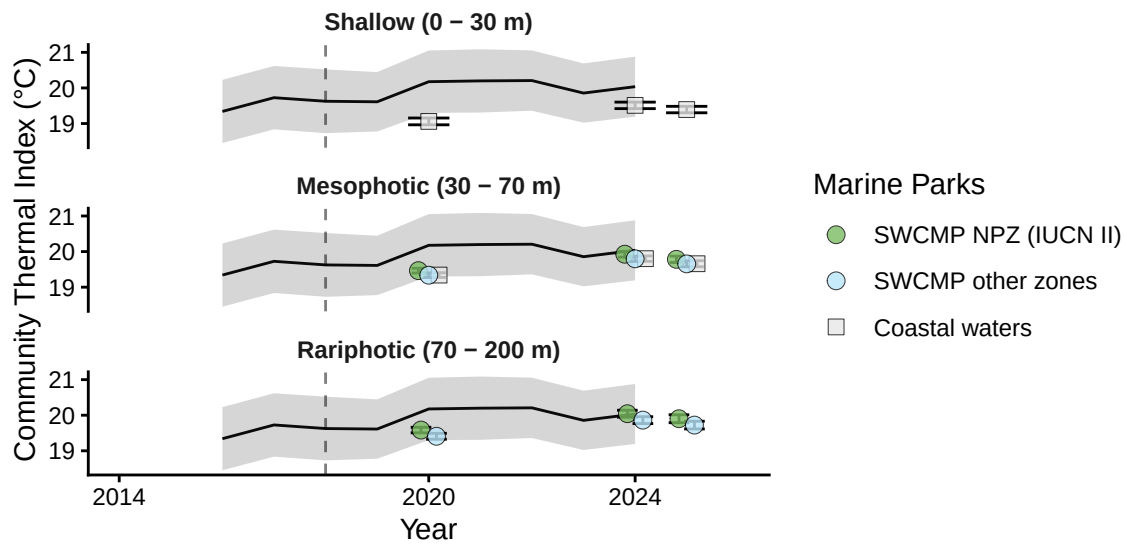


Figure A1.14.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

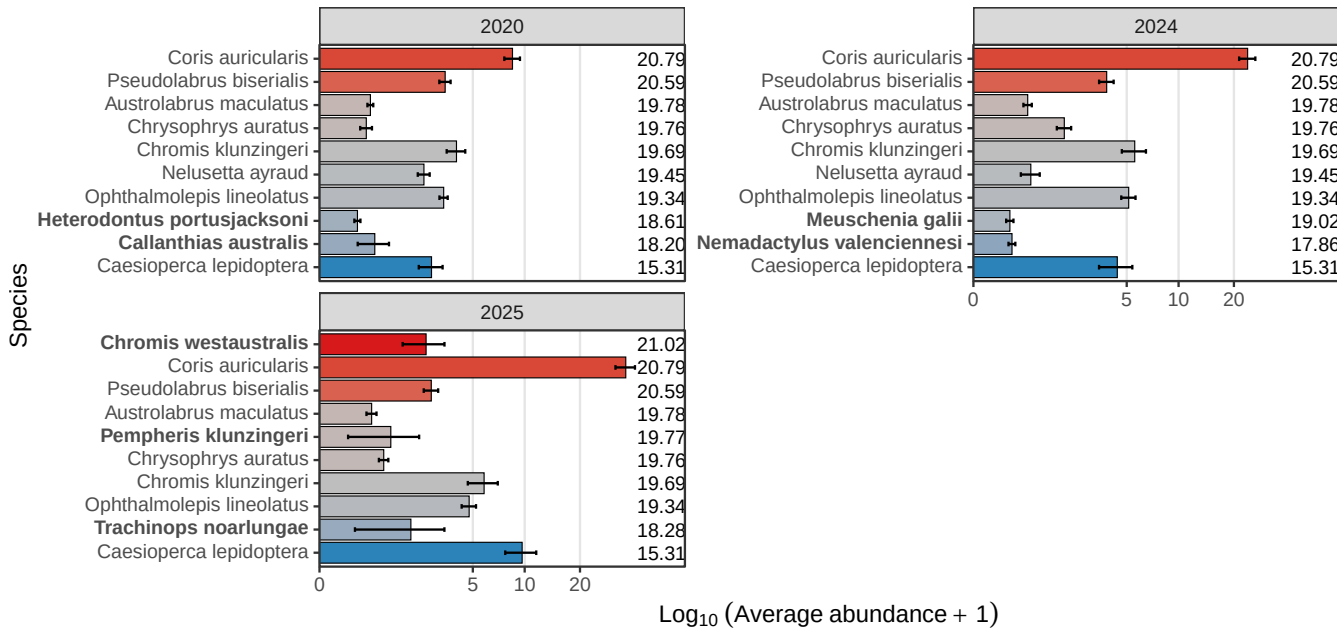


Figure A1.14.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Reef fish thermal index

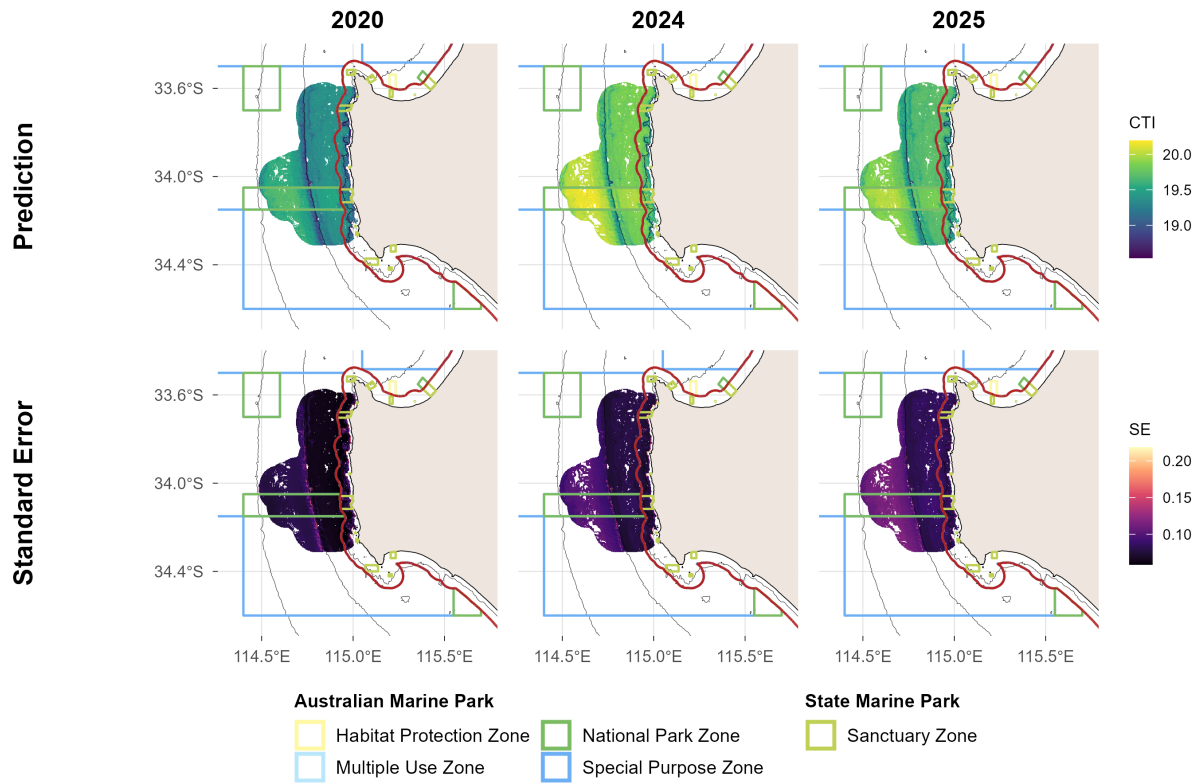


Figure A1.14.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.



## Threatened species

Table A1.2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. \*This species is also listed as ‘Conservation dependent’ on the EPBC Act List of Threatened Fauna. ^This species is also listed as ‘Vulnerable’ on the EPBC Act List of Threatened Fauna.

| Species                        | Common name             | Family         | IUCN category         | Samples | Total MaxN | Years recorded         |
|--------------------------------|-------------------------|----------------|-----------------------|---------|------------|------------------------|
| <i>Epinephelides armatus</i>   | Breaksea Cod            | Serranidae     | Near Threatened       | 122     | 162        | 2020, 2023, 2024, 2025 |
| <i>Bodianus frenchii</i>       | Foxfish                 | Labridae       | Near Threatened       | 99      | 116        | 2020, 2023, 2024, 2025 |
| <i>Sepia apama</i>             | Giant cuttlefish        | Sepiidae       | Near Threatened       | 37      | 44         | 2024, 2025             |
| <i>Squalus acanthias</i>       | Whitespotted Dogfish    | Squalidae      | Vulnerable            | 18      | 28         | 2020                   |
| <i>Achoerodus gouldii</i>      | Western Blue Groper     | Labridae       | Vulnerable            | 14      | 14         | 2020, 2023, 2024, 2025 |
| <i>Carcharhinus brevipinna</i> | Spinner Shark           | Carcharhinidae | Vulnerable            | 13      | 13         | 2020, 2023, 2024, 2025 |
| <i>Carcharhinus obscurus</i>   | Dusky Whaler            | Carcharhinidae | Endangered            | 9       | 12         | 2025                   |
| <i>Carcharhinus plumbeus</i>   | Sandbar Shark           | Carcharhinidae | Endangered            | 7       | 7          | 2020, 2024, 2025       |
| <i>Galeocerdo cuvier</i>       | Tiger Shark             | Carcharhinidae | Near Threatened       | 6       | 6          | 2024, 2025             |
| <i>Carcharhinus limbatus</i>   | Common Blacktip Shark   | Carcharhinidae | Vulnerable            | 4       | 5          | 2020                   |
| <i>Sphyrna zygaena</i> *       | Smooth Hammerhead       | Sphyrnidae     | Vulnerable            | 4       | 4          | 2020, 2024             |
| <i>Galeorhinus galeus</i> *    | School Shark            | Triakidae      | Critically Endangered | 2       | 2          | 2020                   |
| <i>Carcharhinus altimus</i>    | Bignose Shark           | Carcharhinidae | Near Threatened       | 1       | 2          | 2025                   |
| <i>Carcharias taurus</i>       | Greynurse Shark         | Carchariidae   | Critically Endangered | 1       | 1          | 2024                   |
| <i>Rhynchobatus australiae</i> | Whitespotted Guitarfish | Rhinidae       | Critically Endangered | 1       | 1          | 2025                   |
| <i>Sphyrna lewini</i> *        | Scalloped Hammerhead    | Sphyrnidae     | Critically Endangered | 1       | 1          | 2024                   |

## Threatened shark and ray species

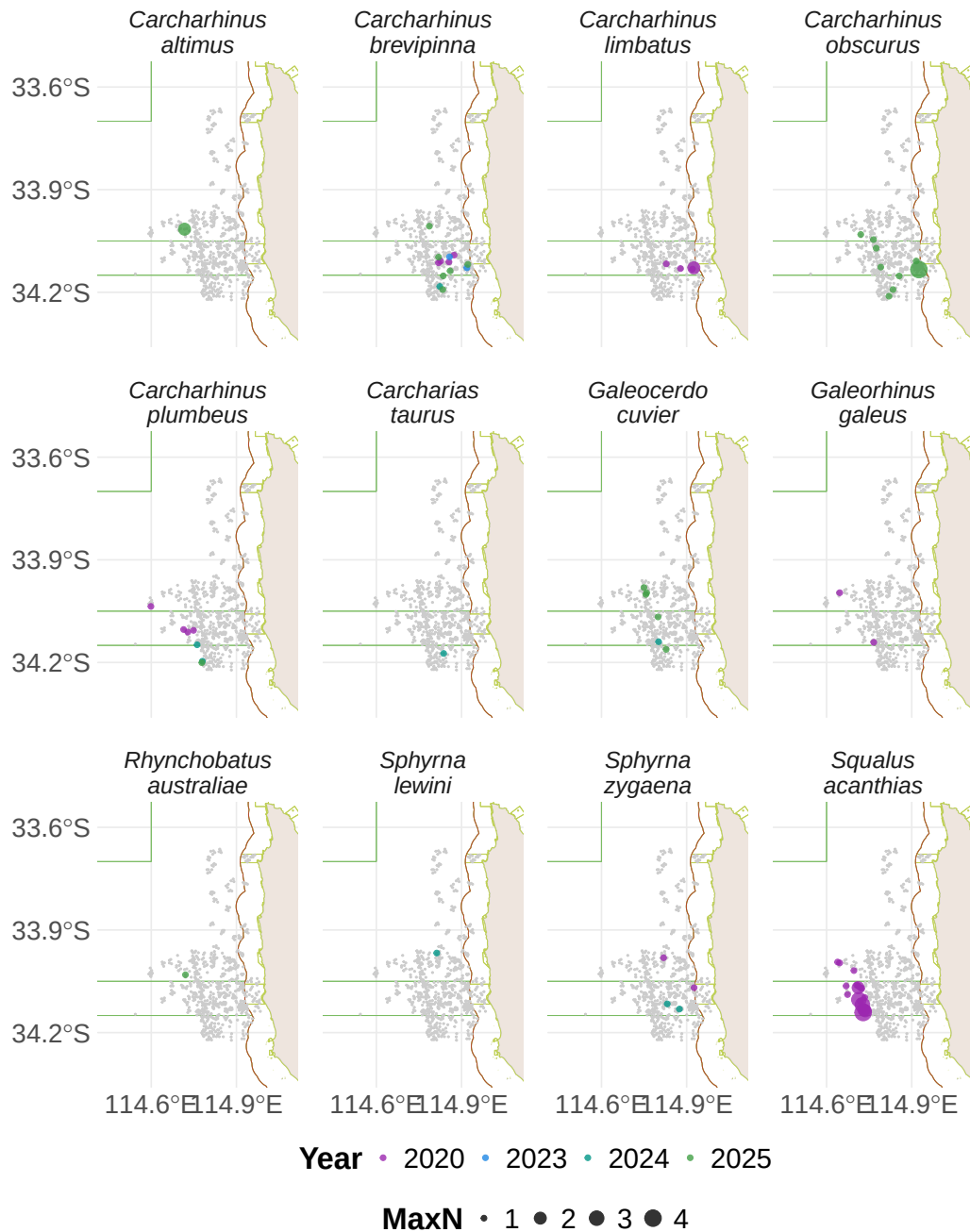


Figure A1.15.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. Commonwealth and coastal marine park boundaries are shown. The red line delimits coastal and Commonwealth waters.

## Other shark and ray species

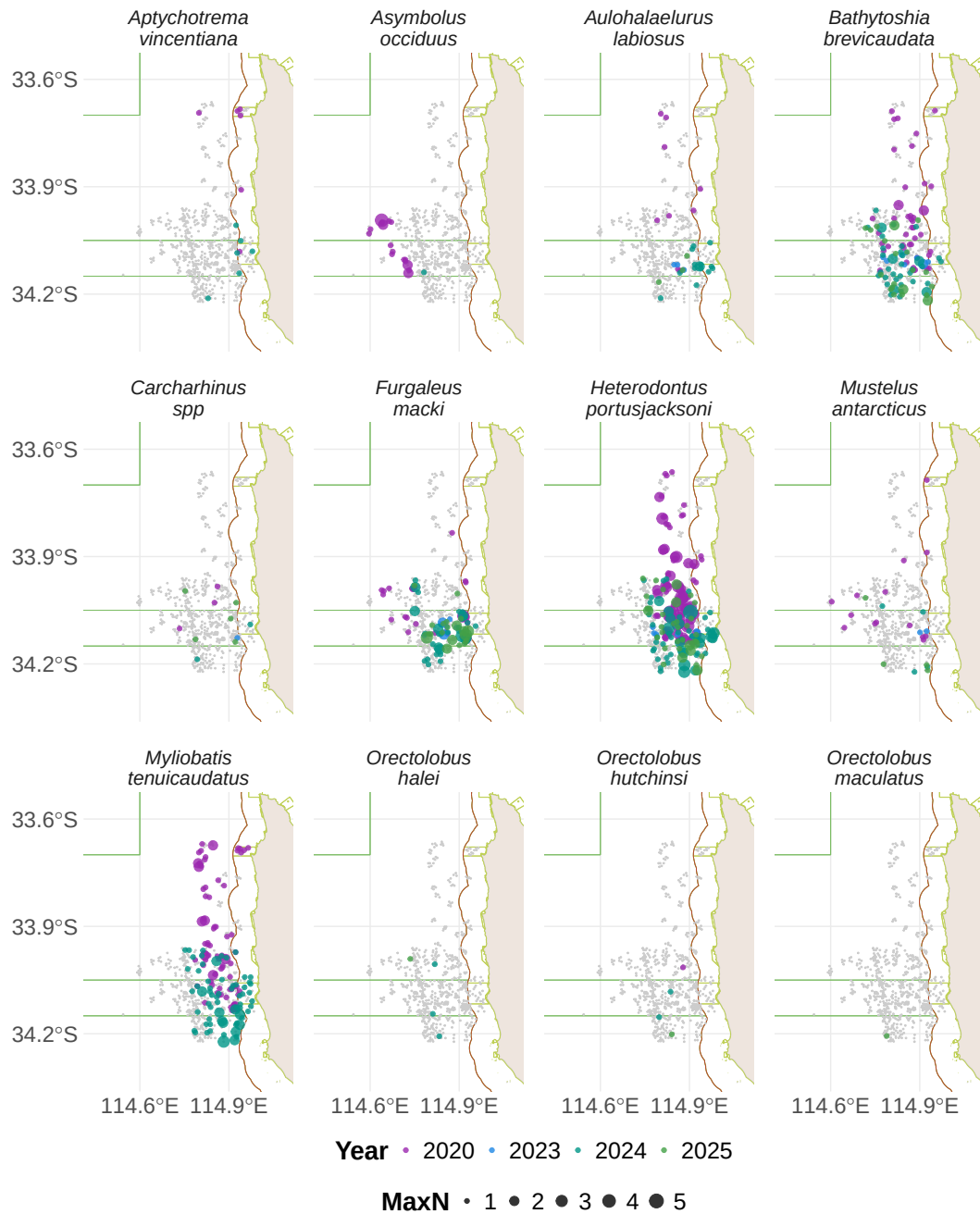


Figure A1.15.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.

## Other shark and ray species

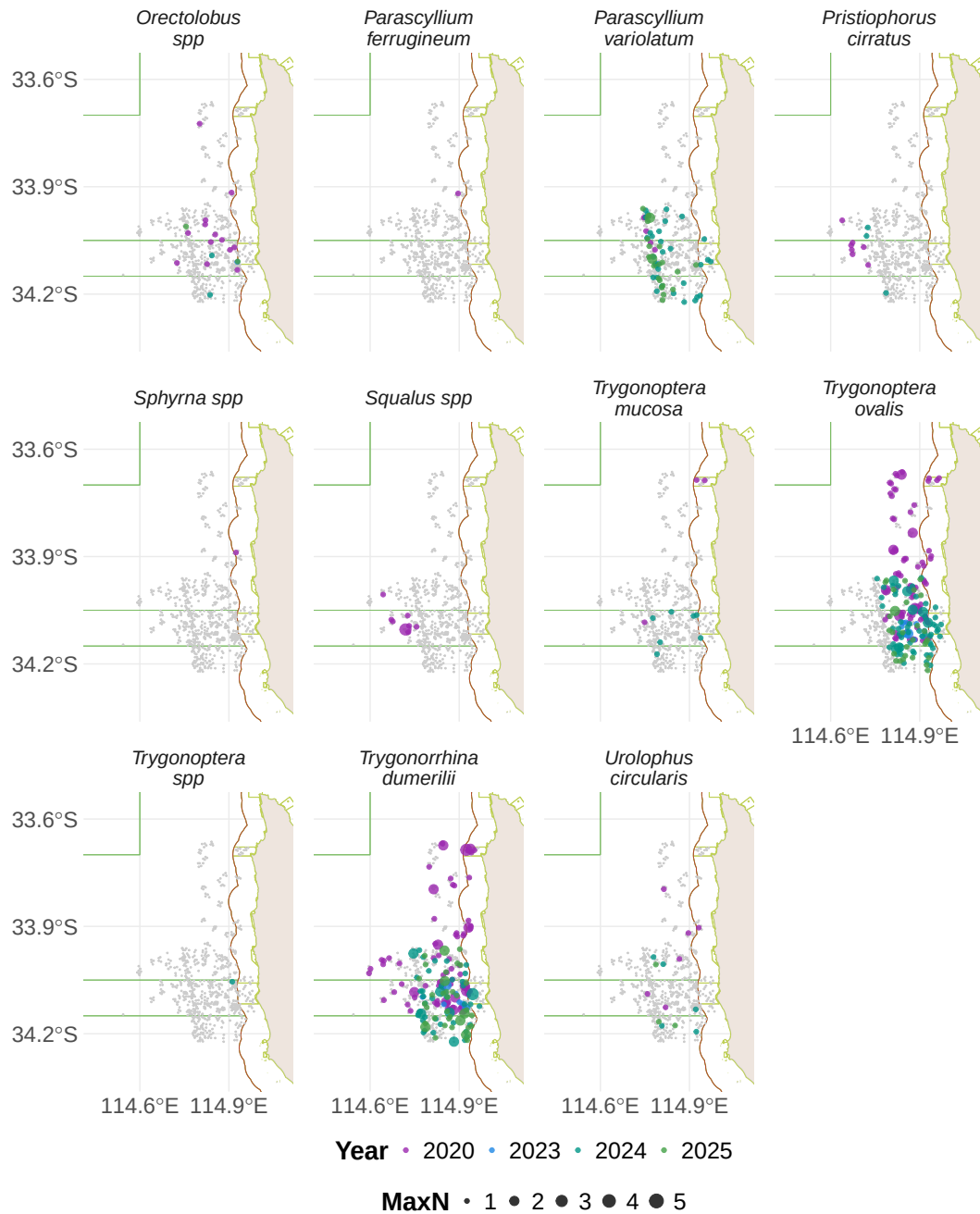


Figure A1.15.3: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery (continued). Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.